ECON1050 Tutorial 10 & 11

Matrix Algebra I & II

Semester 2 2026

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October 12 & 13, 2026

Today's Plan

- This is our second last tutorial. No tutorial in week 13!
- Recap: key ideas from Lecture 10 & 11
- Tutorial questions:
  - Tute 10: Q2, and Q4
  - Tute 11: Q1, Q2, and Q3

Recap: Matrices and notation

Definition

A **matrix** is a rectangular array of numbers.

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

Dimension

A matrix with m rows and n columns has dimension $m \times n$.

Elements

The element a_{ij} is in row i and column j.

Recap: Scalars, vectors, and special matrices

Scalars and vectors

- A **scalar** is a single number.
- A **vector** is a matrix with only one row or only one column.

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \boldsymbol{p}' = \begin{bmatrix} p_1 & p_2 & \cdots & p_n \end{bmatrix}.$$

Special matrices

- A **square matrix** has dimension $n \times n$.
- A **diagonal matrix** has all off-diagonal entries equal to zero.
- An **identity matrix** is a diagonal matrix with ones on the main diagonal.
- A **zero matrix** has all entries equal to zero.



Recap: Equality, addition, and scalar multiplication

Equal matrices

Two matrices are equal only if they have the same dimension and all corresponding entries are identical.

Addition and subtraction

If $\boldsymbol{A}$ and $\boldsymbol{B}$ have the same dimension, then

$$\boldsymbol{A} + \boldsymbol{B} = [a_{ij} + b_{ij}], \qquad \boldsymbol{A} - \boldsymbol{B} = \boldsymbol{A} + (-1)\boldsymbol{B}.$$

Scalar multiplication

For a scalar k,

$$k\boldsymbol{A} = [ka_{ij}].$$

Recap: Properties of matrix addition

Addition properties

For conformable matrices:

$$\boldsymbol{A} + \boldsymbol{B} = \boldsymbol{B} + \boldsymbol{A} \qquad \text{(commutative)}$$

$$\boldsymbol{A} + (\boldsymbol{B} + \boldsymbol{C}) = (\boldsymbol{A} + \boldsymbol{B}) + \boldsymbol{C} \qquad \text{(associative)}$$

$$\boldsymbol{A} + \boldsymbol{0} = \boldsymbol{A}.$$

Scalar multiplication properties

$$k(\boldsymbol{A} + \boldsymbol{B}) = k\boldsymbol{A} + k\boldsymbol{B}, \qquad (k_1 + k_2)\boldsymbol{A} = k_1\boldsymbol{A} + k_2\boldsymbol{A}.$$

Recap: Transpose of a matrix

Definition

The **transpose** of $\boldsymbol{A}$ is denoted by $\boldsymbol{A}'$ or $\boldsymbol{A}^T$. The columns of $\boldsymbol{A}$ become the rows of $\boldsymbol{A}'$.

$$\boldsymbol{A} = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \quad \Rightarrow \quad \boldsymbol{A}' = \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix}.$$

Key properties

$$(\boldsymbol{A}')' = \boldsymbol{A}, \qquad (c\boldsymbol{A})' = c\boldsymbol{A}', \qquad (\boldsymbol{A} + \boldsymbol{B})' = \boldsymbol{A}' + \boldsymbol{B}'.$$

Recap: Matrix multiplication

Conformability

If $\boldsymbol{A}$ is $m \times n$ and $\boldsymbol{B}$ is $n \times p$, then $\boldsymbol{AB}$ is defined and has dimension $m \times p$.

$$\boldsymbol{A}_{m\times n}\boldsymbol{B}_{n\times p} = \boldsymbol{C}_{m\times p}.$$

Element formula

The (i, j) entry of $\boldsymbol{C} = \boldsymbol{AB}$ is

$$c_{ij} = \sum_{r=1}^{n} a_{ir} b_{rj}.$$

This is the inner product of row i of $\boldsymbol{A}$ and column j of $\boldsymbol{B}$.

Recap: Matrix multiplication rules

Important warning

In general,

$$\boldsymbol{AB} \neq \boldsymbol{BA}.$$

Matrix multiplication is not commutative.

Useful properties

$$\boldsymbol{A}(\boldsymbol{BC}) = (\boldsymbol{AB})\boldsymbol{C},$$
$$\boldsymbol{A}(\boldsymbol{B}+\boldsymbol{C}) = \boldsymbol{AB}+\boldsymbol{AC},$$
$$(\boldsymbol{A}+\boldsymbol{B})\boldsymbol{C} = \boldsymbol{AC}+\boldsymbol{BC}.$$
$$(\boldsymbol{AB})' = \boldsymbol{B}'\boldsymbol{A}'.$$

Recap: Powers and special square matrices

Powers

If $\boldsymbol{A}$ is square, then

$$\boldsymbol{A}^2 = \boldsymbol{A}\boldsymbol{A}, \qquad \boldsymbol{A}^3 = \boldsymbol{A}\boldsymbol{A}\boldsymbol{A}.$$

Idempotent matrix

A square matrix $\boldsymbol{A}$ is **idempotent** if

$$\boldsymbol{A}^2 = \boldsymbol{A}.$$

Symmetric and orthogonal matrices

- $\boldsymbol{A}$ is **symmetric** if $\boldsymbol{A} = \boldsymbol{A}'$.
- $\boldsymbol{A}$ is **orthogonal** if $\boldsymbol{A}'\boldsymbol{A} = \boldsymbol{I}$.

Recap: Vectors, inner product, and orthogonality

Inner product

For two n-vectors $\boldsymbol{a}$ and $\boldsymbol{b}$,

$$\boldsymbol{a} \cdot \boldsymbol{b} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = \sum_{i=1}^{n} a_i b_i.$$

Norm

The length or norm of $\boldsymbol{a}$ is

$$\|\boldsymbol{a}\| = \sqrt{\boldsymbol{a} \cdot \boldsymbol{a}} = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}.$$

Orthogonality

Two vectors are orthogonal if

$$\boldsymbol{a} \perp \boldsymbol{b} \quad \Longleftrightarrow \quad \boldsymbol{a} \cdot \boldsymbol{b} = 0.$$

Recap: Systems of linear equations

Matrix form

A system of linear equations can be written as

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},$$

where

- $\boldsymbol{A}$ is the **coefficient matrix**,
- $\boldsymbol{x}$ is the **vector of variables**,
- $\boldsymbol{b}$ is the **vector of constants**.

$$\begin{cases} ax + by = p \\ cx + dy = q \end{cases} \Leftrightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix}$$

Example structure

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} p \\ q \end{bmatrix}.$$

Recap: Elementary row operations

Valid row operations

The following operations do not change the solution set of a system of linear equations:

- Interchange any pair of rows.
- Multiply any row by a nonzero scalar.
- Add any multiple of one row to a different row.

Augmented matrix

For a system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, use

$$[\boldsymbol{A} \mid \boldsymbol{b}].$$

Gaussian elimination applies row operations to this augmented matrix.

$[A|b] \xrightarrow{GE}$ Row-echelon form

Recap: Gaussian elimination

Goal

Convert the coefficient side of the augmented matrix into a simpler form.

Row-echelon form

$$\begin{bmatrix} 1 & a & b & d \\ 0 & 1 & c & e \\ 0 & 0 & 1 & f \end{bmatrix}.$$

Then use backward substitution.

Reduced row-echelon form

$$\begin{bmatrix} 1 & 0 & 0 & g \\ 0 & 1 & 0 & h \\ 0 & 0 & 1 & i \end{bmatrix}.$$

Then the solution is read directly.

Recap: Inverse of a matrix

Definition

The inverse of a square matrix $\boldsymbol{A}$ is denoted by $\boldsymbol{A}^{-1}$ and satisfies

$$\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I} = \boldsymbol{A}^{-1}\boldsymbol{A}.$$

$A \cdot A^{-1} = I$

Solving a system using the inverse

If

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$$

and $\boldsymbol{A}^{-1}$ exists, then

$$\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}.$$

$Ax = b$

$A^{-1}Ax = A^{-1}b$

$I \cdot x = A^{-1}b$

$x = A^{-1}b$

Recap: Properties of inverse matrices

Key facts

- Only square matrices can have inverses.
- Not every square matrix has an inverse.
- If $\boldsymbol{A}^{-1}$ exists, it is unique.
- If $\boldsymbol{A}^{-1}$ exists, then $\boldsymbol{A}$ is **non-singular**.
- A non-singular matrix has linearly independent rows and columns.

Singular matrix

If no inverse exists, $\boldsymbol{A}$ is called **singular**.

Recap: Inverse by Gaussian elimination

Method

For an invertible $n \times n$ matrix $\boldsymbol{A}$:

$$[\boldsymbol{A} \mid \boldsymbol{I}] \quad \longrightarrow \quad [\boldsymbol{I} \mid \boldsymbol{A}^{-1}].$$

Interpretation

The same row operations that transform $\boldsymbol{A}$ into $\boldsymbol{I}$ transform $\boldsymbol{I}$ into $\boldsymbol{A}^{-1}$.

$[A|I] \xrightarrow{GE} [I|A^{-1}]$

Recap: Determinants

Definition

A determinant is a scalar number associated with a square matrix.

$$|\boldsymbol{A}| = \det(\boldsymbol{A}).$$

Second-order determinant

For

$$\boldsymbol{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$|\boldsymbol{A}| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Recap: Determinants and non-singularity

Non-singularity

A square matrix $\boldsymbol{A}$ is non-singular if and only if

$$|\boldsymbol{A}| \neq 0.$$

Singularity

If

$$|\boldsymbol{A}| = 0,$$

then $\boldsymbol{A}$ has no inverse.

Systems of equations

A system $\boldsymbol{Ax} = \boldsymbol{b}$ has a unique solution when $\boldsymbol{A}$ is non-singular.

Recap: Third-order determinants

Expansion by cofactors

For a 3×3 matrix, the determinant can be found by expanding along a row or column.

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \Rightarrow \quad |\boldsymbol{A}| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}.$$

$C_{ij} = (-1)^{i+j}|M_{ij}|$ is Cofactor, where M_{ij} is the minor obtained by deleting row i and column j.

Sarrus' Rule

$$\begin{aligned} |\boldsymbol{A}| = &+ a_{11} \times a_{22} \times a_{33} + a_{12} \times a_{23} \times a_{31} + a_{13} \times a_{21} \times a_{32} \\ &- a_{31} \times a_{22} \times a_{13} - a_{32} \times a_{23} \times a_{11} - a_{33} \times a_{21} \times a_{12} \end{aligned}$$

Recap: Properties of determinants

Key properties

- If a row or column contains only zeros, then $|\boldsymbol{A}| = 0$.
- $|\boldsymbol{A}| = |\boldsymbol{A}'|$.
- If $\boldsymbol{A}$ and $\boldsymbol{B}$ are square matrices of the same order, then

$$|\boldsymbol{AB}| = |\boldsymbol{A}|\,|\boldsymbol{B}|.$$

- Interchanging two rows or two columns changes the sign of the determinant.

Recap: More determinant properties

Scaling and row operations

- Multiplying one row or column by k multiplies the determinant by k.
- If $\boldsymbol{A}$ is $n \times n$, then
$$|k\boldsymbol{A}| = k^n |\boldsymbol{A}|.$$
- Adding a multiple of one row or column to another row or column does not change the determinant.
- If one row or column is a multiple of another, then $|\boldsymbol{A}| = 0$.

Tutorial 10: Q1

Question 1

Evaluate the following integrals using integration by substitution:

$$(1) \quad \int \frac{3x}{(3x^2-4)^2}\,dx, \qquad (2) \quad \int \frac{1}{\sqrt{x}} e^{\sqrt{x}}\,dx.$$

Tutorial Q1: Integration by substitution

Knowledge used

- Revision from integration: use substitution as the reverse of the chain rule.
- If $u = g(x)$ and $du = g'(x)\,dx$, then

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du.$$

Solution

For

$$\int \frac{3x}{(3x^2-4)^2}\,dx,$$

let $u = 3x^2 - 4$, so $du = 6x\,dx$. Hence

$$\int \frac{3x}{(3x^2-4)^2}\,dx = \frac{1}{2}\int u^{-2}\,du = -\frac{1}{2(3x^2-4)} + C.$$

Tutorial Q1: Integration by substitution

Knowledge used

- Revision from integration: use substitution as the reverse of the chain rule.
- If $u = g(x)$ and $du = g'(x)\,dx$, then

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du.$$

Solution

For

$$\int \frac{1}{\sqrt{x}} e^{\sqrt{x}}\,dx,$$

let $u = \sqrt{x}$, so $du = \frac{1}{2\sqrt{x}}dx$. Hence

$$\int \frac{1}{\sqrt{x}} e^{\sqrt{x}}\,dx = 2\int e^{u}\,du = 2e^{\sqrt{x}} + C.$$

Question 2

Use the following matrices:

4 mins

$$\boldsymbol{A} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} -4 \\ 0 \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix},$$

$$\boldsymbol{D} = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix}, \quad \boldsymbol{E} = \begin{bmatrix} 1 & -1 \\ 3 & 0 \\ 2 & 5 \end{bmatrix}, \quad \boldsymbol{F} = \begin{bmatrix} 4 & 1 & -4 \\ 2 & 1 & 6 \\ -1 & 0 & 3 \end{bmatrix}.$$

Calculate the following matrices where possible:
(a) $\boldsymbol{A} + \boldsymbol{B}$; (b) $\boldsymbol{A} - 5\boldsymbol{B}$; (c) $\boldsymbol{B} + 4\boldsymbol{C}$; (d) $\boldsymbol{C}'$; (e) $\boldsymbol{D} + \boldsymbol{E}'$; (f) $(2\boldsymbol{F}')'$.

(b) $A - 5B = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - 5\begin{pmatrix} -4 \\ 0 \end{pmatrix} = \begin{pmatrix} 1+20 \\ 2-0 \end{pmatrix} = \begin{pmatrix} 21 \\ 2 \end{pmatrix}$

(c) $B + 4C = \begin{pmatrix} -4 \\ 0 \end{pmatrix} + 4\begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$

We cannot compute the sum because they have different dimensions

(e) $D + E^T$

$$\boldsymbol{E} = \begin{bmatrix} 1 & -1 \\ 3 & 0 \\ 2 & 5 \end{bmatrix}, \Rightarrow E^T = \begin{pmatrix} 1 & 3 & 2 \\ -1 & 0 & 5 \end{pmatrix}$$

$$D + E^T = \begin{pmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 3 & 2 \\ -1 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 4 \\ 2 & -1 & 6 \end{pmatrix}$$

Tutorial Q2: Matrix addition and scalar multiplication I

Knowledge used

- Matrices can be added only if they have the same dimension.
- Scalar multiplication multiplies every element of a matrix.

Solution

$$\boldsymbol{A}+\boldsymbol{B}=\begin{bmatrix}1\\2\end{bmatrix}+\begin{bmatrix}-4\\0\end{bmatrix}=\begin{bmatrix}-3\\2\end{bmatrix}.$$

$$\boldsymbol{A}-5\boldsymbol{B}=\begin{bmatrix}1\\2\end{bmatrix}-5\begin{bmatrix}-4\\0\end{bmatrix}=\begin{bmatrix}21\\2\end{bmatrix}.$$

Tutorial Q2: Matrix addition and transpose II

Knowledge used

- Check conformability before adding matrices.
- The transpose swaps rows and columns.

Solution

$$\boldsymbol{B} + 4\boldsymbol{C}$$

has no solution because $\boldsymbol{B}$ is 2×1 and $\boldsymbol{C}$ is 2×2.

$$\boldsymbol{C}' = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}' = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}.$$

Tutorial Q2: Transpose operations III

Knowledge used

- Use $(\boldsymbol{A}')' = \boldsymbol{A}$.
- Use $(c\boldsymbol{A})' = c\boldsymbol{A}'$.

Solution

$$\boldsymbol{D} + \boldsymbol{E}' = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 & 2 \\ -1 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 2 & -1 & 6 \end{bmatrix}.$$

$$(2\boldsymbol{F}')' = 2\boldsymbol{F} = \begin{bmatrix} 8 & 2 & -8 \\ 4 & 2 & 12 \\ -2 & 0 & 6 \end{bmatrix}.$$

Tutorial 10: Q3

Question 3

Solve for t:

$$\begin{bmatrix} 2t & 1 \\ -3 & t+1 \end{bmatrix} = 2 \begin{bmatrix} t & 1/2 \\ t+1/2 & -1/2 \end{bmatrix}.$$

Tutorial Q3: Solving from matrix equality

Knowledge used

- Equal matrices must have identical corresponding elements.
- Use scalar multiplication before comparing entries.

Solution

$$2\begin{bmatrix} t & 1/2 \\ t+1/2 & -1/2 \end{bmatrix} = \begin{bmatrix} 2t & 1 \\ 2t+1 & -1 \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} 2t & 1 \\ -3 & t+1 \end{bmatrix} = \begin{bmatrix} 2t & 1 \\ 2t+1 & -1 \end{bmatrix}.$$

Compare corresponding elements:

$$-3 = 2t+1 \quad \Rightarrow \quad t = -2.$$

Also,

$$t+1 = -1 \quad \Rightarrow \quad t = -2.$$

Therefore, $t = -2$.

Tutorial 10: Q4

Question 4

Use the following matrices:

$$\boldsymbol{A} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} -4 \\ 0 \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix},$$

4 matrices

$$\boldsymbol{D} = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix}, \quad \boldsymbol{E} = \begin{bmatrix} 1 & -1 \\ 3 & 0 \\ 2 & 5 \end{bmatrix}, \quad \boldsymbol{F} = \begin{bmatrix} 4 & 1 & -4 \\ 2 & 1 & 6 \\ -1 & 0 & 3 \end{bmatrix}.$$

Compute:

$$\boldsymbol{A}'\boldsymbol{B}, \quad \boldsymbol{A}\boldsymbol{B}', \quad \boldsymbol{A}\boldsymbol{B}, \quad \boldsymbol{A}\boldsymbol{B}'\boldsymbol{C}, \quad \boldsymbol{E}\boldsymbol{D}, \quad (\boldsymbol{E}\boldsymbol{D})', \quad \boldsymbol{C}\boldsymbol{D}\boldsymbol{E}.$$

$AB = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 0 \end{pmatrix}$ $\Rightarrow AB$ is not defined

2×1 2×1

$$AB'C = \begin{pmatrix}1\\2\end{pmatrix}\cdot\begin{pmatrix}-4 & 0\end{pmatrix}\begin{pmatrix}1 & 2\\-1 & 4\end{pmatrix}$$

$2\times1 \quad 1\times2 \quad 2\times2$

$$= \begin{pmatrix}1\times(-4) & 1\times0\\2\times(-4) & 2\times0\end{pmatrix}\begin{pmatrix}1 & 2\\-1 & 4\end{pmatrix}$$

$$= \begin{pmatrix}-4 & 0\\-8 & 0\end{pmatrix}\begin{pmatrix}1 & 2\\-1 & 4\end{pmatrix}$$

$2\times2 \quad 2\times2$

$$= \begin{pmatrix}-4\times1+0\times-1 & -4\times2+0\times4\\-8\times1+0\times2 & -8\times2+0\times4\end{pmatrix}$$

$$= \begin{pmatrix}-4 & -8\\-8 & -16\end{pmatrix}$$

$$\boldsymbol{D}=\begin{bmatrix}1 & 0 & 2\\3 & -1 & 1\end{bmatrix},\quad \boldsymbol{E}=\begin{bmatrix}1 & -1\\3 & 0\\2 & 5\end{bmatrix},$$

$$(ED)'$$

$$ED = \begin{pmatrix}1 & -1\\3 & 0\\2 & 5\end{pmatrix}\cdot\begin{pmatrix}1 & 0 & 2\\3 & -1 & 1\end{pmatrix}$$

$$= \begin{pmatrix}1\times1+(-1)\cdot3 & 1\times0+(-1)\times(-1) & 1\times2+(-1)\times1\\3\times1+0\times3 & 3\times0+0\times(-1) & 3\times2+0\times1\\2\times1+5\times3 & 2\times0+5\times(-1) & 2\times2+5\times\end{pmatrix}$$

$$= \begin{pmatrix}-2 & 1 & 1\\3 & 0 & 6\\17 & -5 & 9\end{pmatrix} \Rightarrow (ED)' = \begin{pmatrix}-2 & 3 & 17\\1 & 0 & -5\\1 & 6 & 9\end{pmatrix}$$

Tutorial Q4: Matrix multiplication I

Knowledge used

- Matrix multiplication requires inner dimensions to match.
- The product of a 1×2 matrix and a 2×1 matrix is 1×1.
- The product of a 2×1 matrix and a 1×2 matrix is 2×2.

Solution

$$\boldsymbol{A}'\boldsymbol{B} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} -4 \\ 0 \end{bmatrix} = -4.$$

$$\boldsymbol{A}\boldsymbol{B}' = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} -4 & 0 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ -8 & 0 \end{bmatrix}.$$

$\boldsymbol{A}\boldsymbol{B}$ is not conformable.

Tutorial Q4: Matrix multiplication II

Knowledge used

- Matrix multiplication is associative:

$$(\boldsymbol{A}\boldsymbol{B}')\boldsymbol{C} = \boldsymbol{A}(\boldsymbol{B}'\boldsymbol{C})$$

whenever the products are defined.

Solution

$$\begin{aligned}\boldsymbol{A}\boldsymbol{B}'\boldsymbol{C} &= \begin{bmatrix}-4 & 0\\ -8 & 0\end{bmatrix}\begin{bmatrix}1 & 2\\ -1 & 4\end{bmatrix}.\\ &= \begin{bmatrix}-4 & -8\\ -8 & -16\end{bmatrix}.\end{aligned}$$

Tutorial Q4: Matrix multiplication III

Knowledge used

- The product of a 3×2 matrix and a 2×3 matrix is 3×3.
- Use row-by-column multiplication for each entry.

Solution

$$\boldsymbol{ED} = \begin{bmatrix} 1 & -1 \\ 3 & 0 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \\ 3 & 0 & 6 \\ 17 & -5 & 9 \end{bmatrix}.$$

$$(\boldsymbol{ED})' = \begin{bmatrix} -2 & 3 & 17 \\ 1 & 0 & -5 \\ 1 & 6 & 9 \end{bmatrix}.$$

Tutorial Q4: Matrix multiplication IV

Knowledge used

- Check dimensions at every step.
- For ***CDE***:

$$(2 \times 2)(2 \times 3)(3 \times 2) \Rightarrow 2 \times 2.$$

Solution

First,

$$\boldsymbol{DE} = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 3 & 0 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 9 \\ 2 & 2 \end{bmatrix}.$$

Then

$$\boldsymbol{CDE} = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 5 & 9 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 9 & 13 \\ 3 & -1 \end{bmatrix}.$$

Key takeaways

- Always check matrix dimensions before adding or multiplying.
- Matrix addition is componentwise, but matrix multiplication is row-by-column.
- In general, $\boldsymbol{AB} \neq \boldsymbol{BA}$.
- Transpose reverses product order: $(\boldsymbol{AB})' = \boldsymbol{B}'\boldsymbol{A}'$.
- Vectors are special matrices, and their inner product is useful in economics.

Tutorial 11: Q1

Question 1

Consider the system

4 m/s

$$2x + y + z = 0, \qquad 4x + 3y + 2z = 2, \qquad 2x - y - 3z = 0.$$

1. Write the system in matrix form.
2. Solve the system using Gaussian elimination.

1. $\begin{cases} 2x + y + z = 0 \\ 4x + 3y + 2z = 2 \\ 2x - y - 3z = 0 \end{cases} \Leftrightarrow \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 2 \\ 2 & -1 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$

$A\,x = b$

2. $Ax = b$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = ?$

$[A \mid b] \xrightarrow{G.E} \text{Row-echelon form}$

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 4 & 3 & 2 & 2 \\ 2 & -1 & -3 & 0 \end{array}\right) \xrightarrow{R_3 : R_3 - R_1} \left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 4 & 3 & 2 & 2 \\ 0 & -2 & -4 & 0 \end{array}\right) \xrightarrow{R_2 : R_2 - 2R_1} \left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & -2 & -4 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & -2 & -4 & 0 \end{array}\right) \xrightarrow{R_3 : R_3 + 2R_2} \left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -4 & 4 \end{array}\right) \quad \text{Row-echelon form}$$

$$\Leftrightarrow (A^* \mid b^*)$$

$$\Leftrightarrow A^* \cdot x = b^*$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -4 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix}$$

$$\Leftrightarrow \begin{cases} 2x + y + z = 0 \\ 0 + y + 0 = 2 \\ 0 + 0 + -4z = 4 \end{cases}$$

$$\Rightarrow \begin{cases} x = -\frac{1}{2} \\ y = 2 \\ z = -1 \end{cases}$$

Tutorial Q1: Matrix form

Knowledge used

- A linear system can be written as $\boldsymbol{Ax} = \boldsymbol{b}$.
- The coefficient matrix contains the coefficients of the variables.

Solution

$$\boldsymbol{Ax} = \boldsymbol{b}$$

where

$$\boldsymbol{A} = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 2 \\ 2 & -1 & -3 \end{bmatrix}, \qquad \boldsymbol{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

Tutorial Q1: Gaussian elimination I

Knowledge used

- Use the augmented matrix $[\boldsymbol{A} \mid \boldsymbol{b}]$.
- Apply elementary row operations to simplify the system.

Solution

$$[\boldsymbol{A} \mid \boldsymbol{b}] = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 4 & 3 & 2 & 2 \\ 2 & -1 & -3 & 0 \end{bmatrix}.$$

First divide row 1 by 2:

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 4 & 3 & 2 & 2 \\ 2 & -1 & -3 & 0 \end{bmatrix}.$$

Then use row 1 to eliminate the first entries in rows 2 and 3: $R_2 \leftarrow R_2 - 4R_1, \quad R_3 \leftarrow R_3 - 2R_1.$

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 2 \\ 0 & -2 & -4 & 0 \end{bmatrix}.$$

Tutorial Q1: Gaussian elimination II

Knowledge used

- Continue row reduction until the system is triangular or reduced.
- Then use backward substitution.

Solution

Use row 2 to eliminate the second entry in row 3: $R_3 \leftarrow R_3 + 2R_2$.

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -4 & 4 \end{bmatrix}.$$

Divide row 3 by -4:

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

Thus $z = -1$, and $y = 2$. Substitute into row 1: $x + \frac{1}{2}(2) + \frac{1}{2}(-1) = 0 \Rightarrow x = -\frac{1}{2}$. Therefore,

$$(x, y, z) = \left(-\frac{1}{2}, 2, -1\right).$$

Tutorial 11: Q2

Question 2

Calculate the inverse of the following matrices, if possible, using Gaussian elimination. Check your result by calculating $\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$.

5 mins

(a) $\begin{bmatrix} 5 & 2 \\ 1 & -1 \end{bmatrix}$, (b) $\begin{bmatrix} 5 & 2 \\ 10 & 4 \end{bmatrix}$,

(c) $\begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix}$.

$|A|$ ① $|A| = 0 \Rightarrow A$ has no inverse
② $|A| \neq 0 \Rightarrow A$ does have an inverse A^{-1}

(b) $\begin{pmatrix} 5 & 2 \\ 10 & 4 \end{pmatrix}$: $\begin{vmatrix} 5 & 2 \\ 10 & 4 \end{vmatrix} = 20 - 20 = 0 \Rightarrow A$ has no inverse

(c) $\begin{vmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{vmatrix} = -2 \neq 0 \Rightarrow A$ has an inverse A^{-1}

Augmented matrix $[A \mid I] \xrightarrow{GE} [I \mid \underline{A^{-1}}]$

Inverse of A

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 4 & -2 & 1 & 0 & 1 & 0 \\ 1 & 2 & -10 & 0 & 0 & 1 \end{array}\right) \begin{matrix} R_2 - 4R_1 \\ R_3 - R_1 \end{matrix} \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & -2 & 9 & -4 & 1 & 0 \\ 0 & 2 & -8 & -1 & 0 & 1 \end{array}\right) \xrightarrow{R_3 + R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & -2 & 9 & -4 & 1 & 0 \\ 0 & 0 & 1 & -5 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & -2 & 9 & -4 & 1 & 0 \\ 0 & 0 & 1 & -5 & 1 & 1 \end{array}\right) \xrightarrow{R_2 - 9R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & -2 & 0 & 41 & -8 & -9 \\ 0 & 0 & 1 & -5 & 1 & 1 \end{array}\right) \xrightarrow{-\frac{R_2}{2}} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -\frac{41}{2} & 4 & \frac{9}{2} \\ 0 & 0 & 1 & -5 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -\frac{41}{2} & 4 & \frac{9}{2} \\ 0 & 0 & 1 & -5 & 1 & 1 \end{array}\right) \xrightarrow{R_1 + 2R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -9 & 2 & 2 \\ 0 & 1 & 0 & -\frac{41}{2} & 4 & \frac{9}{2} \\ 0 & 0 & 1 & -5 & 1 & 1 \end{array}\right)$$

$\Leftrightarrow (I \mid A^{-1})$

$$\Rightarrow A^{-1} = \begin{pmatrix} -9 & 2 & 2 \\ -\frac{41}{2} & 4 & \frac{9}{2} \\ -5 & 1 & 1 \end{pmatrix}$$

$A^{-1} \cdot A = \underline{I}$

Tutorial Q2(a): Inverse by Gaussian elimination

Knowledge used

- To find $\boldsymbol{A}^{-1}$, row-reduce $[\boldsymbol{A} \mid \boldsymbol{I}]$ into $[\boldsymbol{I} \mid \boldsymbol{A}^{-1}]$.
- Verify by checking $\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I}$.

Solution

For

$$\boldsymbol{A} = \begin{bmatrix} 5 & 2 \\ 1 & -1 \end{bmatrix},$$

row-reducing

$$\left[\begin{array}{cc|cc} 5 & 2 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{array}\right]$$

gives

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{7} & \frac{2}{7} \\ 0 & 1 & \frac{1}{7} & -\frac{5}{7} \end{array}\right].$$

Therefore,

$$\boldsymbol{A}^{-1} = \begin{bmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{1}{7} & -\frac{5}{7} \end{bmatrix}.$$

Tutorial Q2(b): Singular matrix

Knowledge used

- A square matrix has no inverse if its rows or columns are linearly dependent.
- Equivalently, a matrix is singular if its determinant is zero.

Solution

For $\boldsymbol{A} = \begin{bmatrix} 5 & 2 \\ 10 & 4 \end{bmatrix}$, the augmented matrix is

$$\left[\begin{array}{cc|cc} 5 & 2 & 1 & 0 \\ 10 & 4 & 0 & 1 \end{array}\right].$$

Apply $R_2 \leftarrow R_2 - 2R_1$. Then row 2 on the coefficient side becomes zero:

$$\left[\begin{array}{cc|cc} 5 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right].$$

So the rows are linearly dependent. Therefore, $\boldsymbol{A}$ is singular and has no inverse.

Tutorial Q2(c): Inverse of a 3×3 matrix

Knowledge used

- Use the same Gaussian elimination method for 3×3 matrices.
- Row-reduce $[\boldsymbol{A} \mid \boldsymbol{I}]$ to $[\boldsymbol{I} \mid \boldsymbol{A}^{-1}]$.

Solution

For

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix},$$

Gaussian elimination gives

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -9 & 2 & 2 \\ 0 & 1 & 0 & -\frac{41}{2} & 4 & \frac{9}{2} \\ 0 & 0 & 1 & -5 & 1 & 1 \end{array}\right].$$

Thus

$$\boldsymbol{A}^{-1} = \begin{bmatrix} -9 & 2 & 2 \\ -\frac{41}{2} & 4 & \frac{9}{2} \\ -5 & 1 & 1 \end{bmatrix}.$$

Tutorial 11: Q3

Question 3

Find the following determinants:

4 mins

$$(1)\ \begin{vmatrix} 5 & 2 \\ 1 & -1 \end{vmatrix}, \qquad (2)\ \begin{vmatrix} 5 & 2 \\ 10 & 4 \end{vmatrix},$$

$$(3)\ \begin{vmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{vmatrix}, \qquad (4)\ \begin{vmatrix} 2 & 1 & 1 \\ 4 & 3 & 2 \\ 2 & -1 & -3 \end{vmatrix}.$$

Expansion by cofactors

For a 3×3 matrix, the determinant can be found by expanding along a row or column.

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \Rightarrow \quad |\boldsymbol{A}| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}.$$

$C_{ij} = (-1)^{i+j}|M_{ij}|$ is Cofactor, where M_{ij} is the minor obtained by deleting row i and column j.

Sarrus' Rule

$$|\boldsymbol{A}| = + a_{11} \times a_{22} \times a_{33} + a_{12} \times a_{23} \times a_{31} + a_{13} \times a_{21} \times a_{32} - a_{31} \times a_{22} \times a_{13} - a_{32} \times a_{23} \times a_{11} - a_{33} \times a_{21} \times a_{12}$$

(2) $\begin{vmatrix} 5 & 2 \\ 10 & 4 \end{vmatrix} = 5\times4 - 2\times10 = 20 - 20 = 0$

(3) $\begin{vmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{vmatrix} = 1\times(-2)\times(-10) + 0\times1\times1 + (-2)\times4\times2 - 1\times(-2)\times(-2) - 4\times0\times(-10) - 1\times2\times1$

$= 20 + 0 - 16 - 4 - 0 - 2$

$= -$

Tutorial Q3: Determinants I

Knowledge used

- For a 2×2 matrix,
$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$
- If a determinant is zero, the matrix is singular.

Solution

$$\begin{vmatrix} 5 & 2 \\ 1 & -1 \end{vmatrix} = 5(-1) - 2(1) = -7.$$

$$\begin{vmatrix} 5 & 2 \\ 10 & 4 \end{vmatrix} = 5(4) - 10(2) = 0.$$

Tutorial Q3: Determinants II

Knowledge used

- For 3×3 matrices, use cofactor expansion.
- Expanding along a row with zeros can reduce calculation.

Solution

$$\begin{vmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{vmatrix}$$

Expand along row 1:

$$= 1\left[(-2)(-10) - 2(1)\right] + (-2)\left[4(2) - 1(-2)\right] = 18 - 20 = -2.$$

$$\begin{vmatrix} 2 & 1 & 1 \\ 4 & 3 & 2 \\ 2 & -1 & -3 \end{vmatrix}$$

Expand along row 1:

$$= 2[3(-3) - 2(-1)] - 1[4(-3) - 2(2)] + 1[4(-1) - 2(3)] = -14 + 16 - 10 = -8.$$

Tutorial 11: Q4

Question 4

A factory produces a slalom water ski that requires 3 labour-hours in the assembly department and 1 labour-hour in the finishing department. Assembly personnel receive \$19 per hour and finishing personnel receive \$16 per hour.

1. Write a matrix or vector product that would give the total labour cost per ski, then evaluate.
2. The factory also produces a trick water ski that requires 5 labour-hours in assembly and 1.5 labour-hours for finishing. Write a product of vectors that will give the total labour cost for this ski.
3. Labour-hour information can be combined into a single matrix $\boldsymbol{L} = \begin{bmatrix} 5 & 1.5 \\ 3 & 1 \end{bmatrix}$. Now suppose the company has two manufacturing plants, one in Sydney and one in Brisbane and that their hourly rates for each department are given by $\boldsymbol{H} = \begin{bmatrix} \$22 & \$19 \\ \$18 & \$16 \end{bmatrix}$ where the assembly and finishing departments are the rows and Sydney and Brisbane are the columns.
   Identify a matrix product that would yield the total labour cost for each type of ski and evaluate.

Tutorial Q4: Labour costs

Knowledge used

- Use matrix multiplication to combine quantities and unit costs.
- A row vector times a column vector gives a scalar product.
- The order of multiplication matters for interpretation.

Solution

Total labour cost per slalom ski:

$$\begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} 19 \\ 16 \end{bmatrix} = 3(19) + 1(16) = 73.$$

Total labour cost per trick ski:

$$\begin{bmatrix} 5 & 1.5 \end{bmatrix} \begin{bmatrix} 19 \\ 16 \end{bmatrix} = 5(19) + 1.5(16) = 119.$$

Tutorial Q4: Matrix product interpretation

Knowledge used

- Matrix multiplication can be conformable but not economically meaningful.
- Always check whether the resulting entries have the intended interpretation.

Solution

$$\boldsymbol{LH} = \begin{bmatrix} 5 & 1.5 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 22 & 19 \\ 18 & 16 \end{bmatrix} = \begin{bmatrix} 137 & 119 \\ 84 & 73 \end{bmatrix}.$$

This product gives total labour costs by ski type and plant.
However,

$$\boldsymbol{HL} = \begin{bmatrix} 22 & 19 \\ 18 & 16 \end{bmatrix} \begin{bmatrix} 5 & 1.5 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 167 & 52 \\ 138 & 43 \end{bmatrix}.$$

Although this product is conformable, it does not relate costs to the correct skis or plants in a meaningful way.

Key takeaways

- A linear system can be written as $\boldsymbol{Ax} = \boldsymbol{b}$.
- Gaussian elimination uses elementary row operations to solve systems.
- If $\boldsymbol{A}^{-1}$ exists, then $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$.
- A square matrix is non-singular if and only if its determinant is nonzero.
- Cofactor expansion and determinant properties help identify whether inverses exist.